RAMCO INSTITUTE OF TECHNOLOGY

Department of Electrical and Electronics Engineering

Academic Year: 2019 – 2020 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE
Course Code & Title: EE8403 Measurements and Instrumentation
Name of the Faculty member: S. Sharmila Kumari
****

ACTIVITY

Topic: Errors in Measurement

Goal (Learning Outcome):

- ❖ The students will be able to explain various types of errors present in measurements - (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: One minute paper
Justification for Choosing the Activity for the topic:

1. Explain various types of errors

   The topic Errors in Measurements has 9 types of errors in instruments. Since each type has its own definition, limitation and examples. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the each type of errors which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 4 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

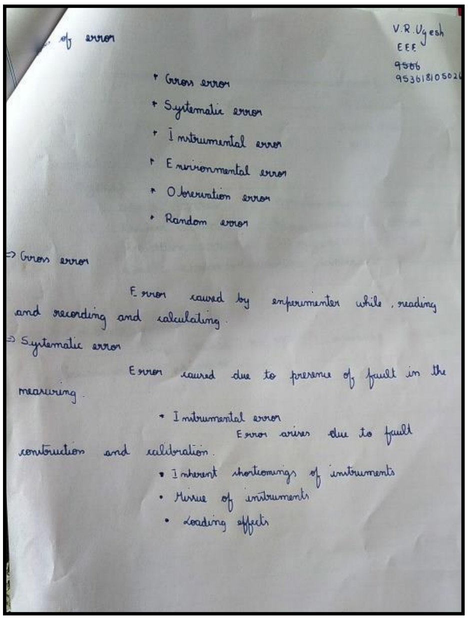
Total Strength is 29

Reporter: Myself

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about various types of errors in measurements concept for 2 minutes.
- ✓ Then I told them to write as much as they can within a short period of time (1 minute)
- ✓ Finally, I collected the papers from each column (1 minute)

Samples:


Types of error

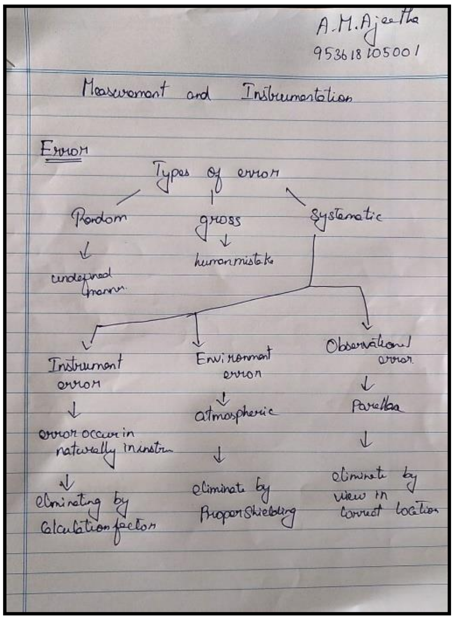
V.R Ugesh
EEE
9586
95361810502

• Gross error
• Systematic error
• Instrumental error
• Environmental error
• Observation error
• Random error

⇒ Gross error
Error caused by experimenter while reading and recording and calculating

⇒ Systematic error
Error caused due to presence of fault in the measuring.

• Instrumental error
Error arises due to fault construction and calibration.
• Inherent shortcomings of instruments
• Misuse of instruments
• Loading effects


A.M.Ajeetha
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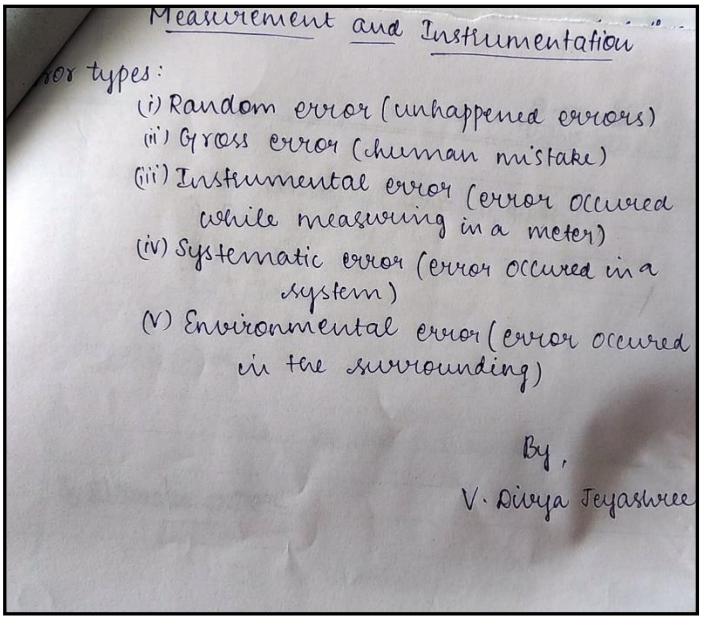
Measurement and Instrumentation

Error

Types of error
Random → undefined manner
gross → human mistake
Systematic
Instrument error → error occur in naturally instrument → eliminating by calculation factor
Environment error → atmospheric → eliminate by proper shielding
Observational error → Parallax → eliminate by view in correct location


Measurement and Instrumentation

Error types:
(i) Random error (unhappened errors)
(ii) Gross error (human mistake)
(iii) Instrumental error (error occured while measuring in a meter)
(iv) Systematic error (error occured in a system)
(V) Environmental error (error occured in the surrounding)

By,
V. Divya Jeyashree

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain various types of errors present in measurements.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 50% of students answered.**

Reflective Critique:

1. **Pre-implementation Reflection :**

- **Benefits:**
  - ➢ Students can able to attend the question even in the questions are in indirect form.

- Students can able to explain the concepts in examination without any confusion.

- **Challenges:**
  - In the class mostly boys hesitate to answer the questions.
  - Time utilization for conducting activity.

2. **Post-implementation Reflection :**

- **Benefits:**
  - Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- **Challenges:**
  - Slow learners were not able to understand some topics during discussion hours.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
C212.1	3	3	1	-	1	-	-	1	-	1	-	1

References:

1. A.K. Sawhney, 'A Course in Electrical & Electronic Measurements & Instrumentation', Dhanpat Rai and Co, 2010.
2. H.S. Kalsi, 'Electronic Instrumentation', McGraw Hill, III Edition 2010.

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ACTIVITY

Topic: Principle and types of analog meter

Goal (Learning Outcome):

- ❖ The students will be able to explain the principle and types of analog meter - (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: Demonstration
Justification for Choosing the Activity for the topic:

1. Explain the principle and types of analog meter
   The topic Principle and types of analog meter has 7 types of instruments. Since each type has its own principle of operation, torque equation and construction. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the each type of meters which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 29,

Photographer: one student - Mr. V R Ugesh (interested in photography)

Reporter: Myself

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about various types of meters in measurements concept for 2 minutes.
- ✓ Then I told them to I told them to Pair with their neighbours and discuss about the construction parts of instruments for another 1 minute.

- ✓ Finally, I show the PMMC and PMMI ammeter for each row (3 minutes)

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students can able to explain the Analog meters (MC, MI, and Electrodynamometer) construction and working principle.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 80% of students answered.**

Reflective Critique:

3. **Pre-implementation Reflection :**

- **Benefits:**

- ➢ Students can able to attend the question even in the questions are in indirect form.
- ➢ Students can able to explain the concepts in examination without any confusion.

- **Challenges:**
  - ➢ In the class mostly girls hesitate to answer the questions.
  - ➢ Time utilization for conducting activity.

4. Post-implementation Reflection :

- **Benefits:**
  - ➢ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- **Challenges:**
  - ➢ Slow learners were not able to understand some topics during discussion hours.

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2
C212. 1	3	3	1	-	1	-	-	1	-	1	-	1

References:

1. A.K. Sawhney, ‘A Course in Electrical & Electronic Measurements & Instrumentation’, Dhanpat Rai and Co, 2010.
2. https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/172-demonstrations

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ACTIVITY

Topic: Wattmeter and Energy meter

Goal (Learning Outcome):

- ❖ The students will be able to explain the working principle of Wattmeter and Energymeter various types of errors present in measurements - (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: Demonstration
Justification for Choosing the Activity for the topic:

2. Explain Wattmeter and Energy meter

After teaching the concept, I thought of conducting this activity for making the students to give the difference between the Wattmeter and Energy meter which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 29,

Photographer: one student - Mr. V R Ugesh (interested in photography)

Reporter: Myself

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about various types of meters in measurements concept for 2 minutes.
- ✓ Then I told them to I told them to pair with their neighbours and discuss about the construction parts of instruments for another 1 minute.

- ✓ Finally, I show the Electrodynamometer type wattmeter and induction type energy meter for each row (3 minutes)

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.

- ✓ While conducting the activity, I understood that the students can able to explain the analog meters (Electrodynamometer and Wattmeter) construction and working principle.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 80% of students answered.**

Reflective Critique:

5. **Pre-implementation Reflection :**

- **Benefits:**
  - ➢ Students can able to attend the question even in the questions are in indirect form.
  - ➢ Students can able to explain the concepts in examination without any confusion.
- **Challenges:**
  - ➢ In the class mostly girls hesitate to answer the questions.
  - ➢ Time utilization for conducting activity.

6. **Post-implementation Reflection :**

- **Benefits:**
  - ➢ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- **Challenges:**
  - ➢ Slow learners were not able to understand some topics during discussion hours.

CO	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO1 0**	**PO1 1**	**PO1 2**
C212. 2	3	2	2	-	1	-	-	1	-	1	-	1

References:

1. A.K. Sawhney, ‘A Course in Electrical & Electronic Measurements & Instrumentation’, Dhanpat Rai and Co, 2010.
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ACTIVITY

Topic: Electrostatic and electromagnetic interference

Goal (Learning Outcome):

- The students will be able to compare Electrostatic and electromagnetic interference - (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: Sage and Scribe
Justification for Choosing the Activity for the topic:

1. Differentiate Electrostatic and electromagnetic interference
   After teaching the concept, I thought of conducting this activity for making the students to give the difference between the Electrostatic and electromagnetic interference which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept, give students one or two minutes to think about the topic without writing anything and tell pair with neighbour 2 in desk.

Total Strength is 29, Number of Pairs – 14

Photographer: one student - Mr. V R Ugesh (interested in photography)

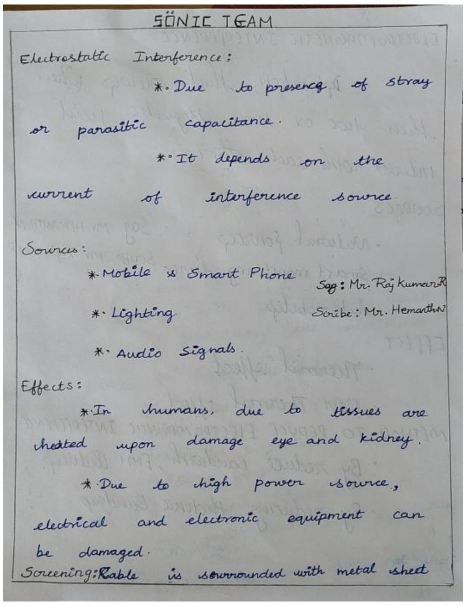
Reporter: Myself

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about types of interference in measurements concept for 2 minutes.
- ✓ Then I told them to I told them to pair with their neighbours and play one person as sage role and other person as scribe role

✓ Scribe take a notes about topic with the help of sage instruction (3 minutes)

Samples:



SÖNIC TEAM

Electrostatic Interference:

*. Due to presencq of stray or parasitic capacitance.

*. It depends on the current of interference source

Sources:

*. Mobile & Smart Phone

*. Lighting

*. Audio Signals.

Sag: Mr. Raj Kumar R

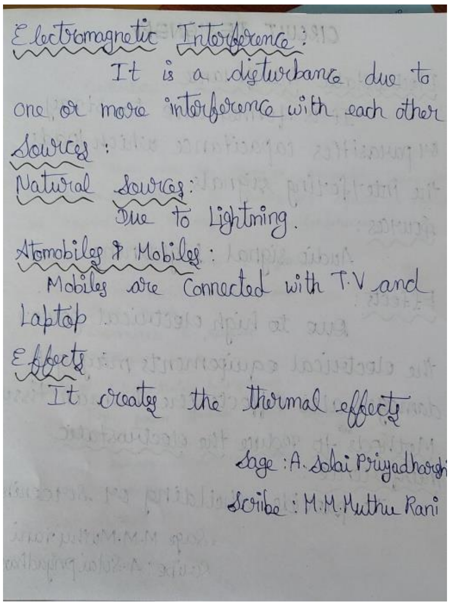
Scribe: Mr. Hemanth N

Effects:

*. In humans, due to tissues are heated upon damage eye and kidney.

*. Due to high power source, electrical and electronic equipment can be damaged.

Screening: Kable is surrounded with metal sheet



Electromagnetic Interference:

It is a disturbance due to one or more interference with each other

Sources:

Natural Sources:

Due to Lightning.

Atomobiles & Mobiles:

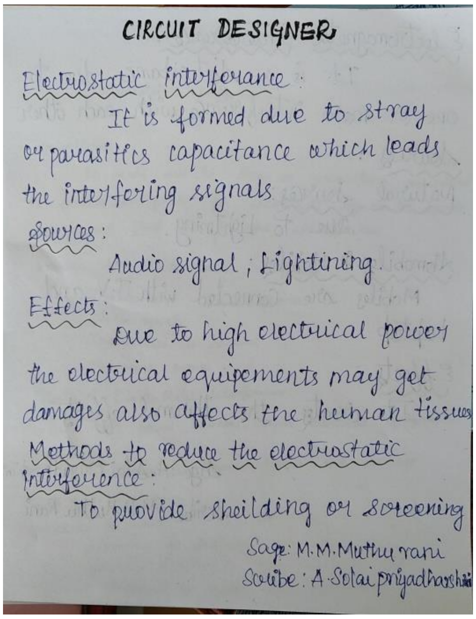
Mobiles are Connected with T.V and Laptop

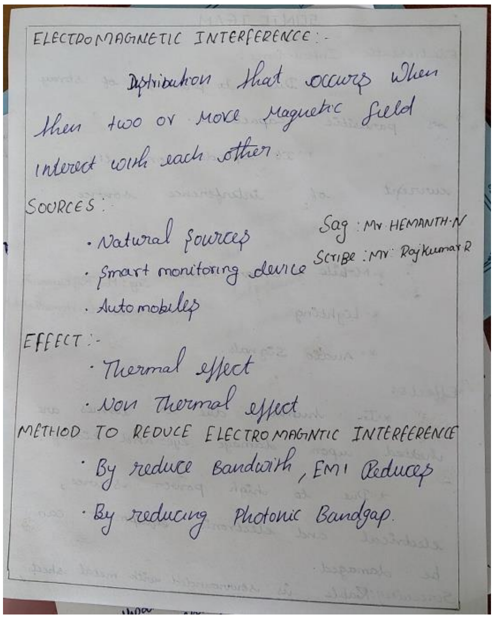
Effects:

It creates the thermal effects

Sage: A. Solai Priyadharshi

Scribe: M.M. Muthu Rani





Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students can able to explain the difference between Electrostatic and electromagnetic interference
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 60% of students answered.**

Reflective Critique:

7. Pre-implementation Reflection :

- **Benefits:**
  - ➢ Students can able to attend the question even in the questions are in indirect form.
  - ➢ Students can able to explain the concepts in examination without any confusion.
- **Challenges:**
  - ➢ In the class mostly girls hesitate to answer the questions.
  - ➢ Time utilization for conducting activity.

8. Post-implementation Reflection :

- **Benefits:**
  - ➢ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- **Challenges:**
  - ➢ Slow learners were not able to understand some topics during discussion hours.

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2
C212. 3	2	2	3	-	1	-	-	-	-	1	-	1

References:

2. A.K. Sawhney, 'A Course in Electrical & Electronic Measurements & Instrumentation', Dhanpat Rai and Co, 2010.
3. https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/172-Sage and scribe

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ACTIVITY

Topic: Cathode Ray Tube (CRT) display

Goal (Learning Outcome):

❖ The students will be able to explain the parts of CRT display – (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: Concept map

Justification for Choosing the Activity for the topic:

1. Cathode Ray Tube (CRT) display

   After teaching the concept, I thought of conducting this activity for making the students able to explain the parts of CRT display which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept, the students were made to pair with their neighbours

Photographer: Myself

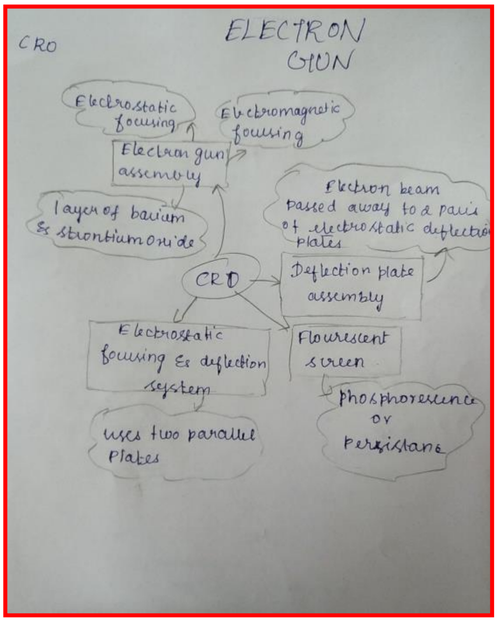
Reporter: Myself

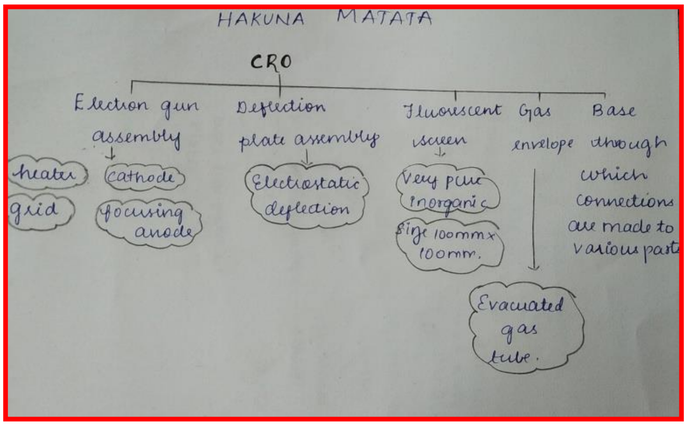
At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about the parts of CRT display for 2 minute.
- ✓ Then I told them to Pair with their neighbours and discuss about the concepts for another 1 minute.
- ✓ Finally I told them to design Concept map and submit within 3 minutes.

A ***view of implementation of the Activity***

Samples:





Reflective Critique:

- **Benefits:**
    - ➢ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- **Challenges:**
    - ➢ Slow learners were not able to understand some topics during discussion hours.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C212.4	3	2	1	-	-	-	-	1	-	1	-	1

References:

1. https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27- teaching/184-visual-modeling-concept-maps